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C:\Windows\System32>cd C:\Users\lohit\OneDrive\Documents\MongoDB\mongosh-2.10.0-win32-x64\bin

C:\Users\lohit\OneDrive\Documents\MongoDB\mongosh-2.10.0-win32-x64\bin>mongosh "mongodb+srv://cluster0.eay9vj1.mongodb.net/" --apiVersion 1 --username <db_username> --password Khd0Zk187Jces0
The system cannot find the file specified.

C:\Users\lohit\OneDrive\Documents\MongoDB\mongosh-2.10.0-win32-x64\bin>mongosh "mongodb+srv://cluster0.eay9vj1.mongodb.net/" --apiVersion 1 --username <lohitshurya_db_user>
The syntax of the command is incorrect.

C:\Users\lohit\OneDrive\Documents\MongoDB\mongosh-2.10.0-win32-x64\bin>mongosh "mongodb+srv://cluster0.eay9vj1.mongodb.net/" --apiVersion 1 --username lohitshurya_db_user
Enter password: ****
Current Mongosh Log ID: 6a8677cd3c882e93d293a163
Connecting to:
  mongodb+srv://<credentials>@cluster0.eay9vj1.mongodb.net/?appName=mongosh+2.10.0
Using MongoDB:
  8.0.29 (API Version 1)
Using Mongosh:
  2.10.0
Atlas atlas-4upy6x-shard-0 [primary] test> use schoolDB;
switched to db schoolDB
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.insertOne({ name: "Alice", age: 21, major: "Computer Science" })
{
  acknowledged: true,
  insertedId: ObjectId('6a8679983c882e93d293a164')
}
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.insertMany([
  { name: "Bob", age: 22, major: "Mathematics" },
  { name: "Charlie", age: 23, major: "Physics" },
  { name: "David", age: 21, major: "Computer Science" },
  { name: "Eve", age: 22, major: "Mathematics" }
]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a867ad63c882e93d293a165'),
    '1': ObjectId('6a867ad63c882e93d293a166'),
    '2': ObjectId('6a867ad63c882e93d293a167'),
    '3': ObjectId('6a867ad63c882e93d293a168')
  }
}
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.find({ major: "Computer Science" });
[
  {
    _id: ObjectId('6a8679983c882e93d293a164'),
    name: 'Alice',
    age: 21,
    major: 'Computer Science'
  },
]

```

```

C:\Users\lohit\OneDrive\Documents\MongoDB\mongosh-2.10.0-win32-x64\bin>mongosh "mongodb+srv://cluster0.eay9vj1.mongodb.net/"
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.find({ major: "Computer Science" });
[
  {
    _id: ObjectId('6a8679983c882e93d293a164'),
    name: 'Alice',
    age: 21,
    major: 'Computer Science'
  },
  {
    _id: ObjectId('6a867ad63c882e93d293a167'),
    name: 'David',
    age: 21,
    major: 'Computer Science'
  }
]
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.updateOne({ name: "Alice" }, { $set: { age: 22 } });
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.deleteOne({ name: "Alice" })
{
  acknowledged: true,
  deletedCount: 1
}
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.insertOne({ name: "Alice", age: 21, major: "Computer Science" });
{
  acknowledged: true,
  insertedId: ObjectId('6a867cb73c882e93d293a169')
}
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.createIndex({ major: 1 });
major_1
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.getIndexes();
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { major: 1 }, name: 'major_1' }
]
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.createIndex({ major: 1, age: -1 });
major_1_age_-1
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.dropIndex("major_1");
{
  nIndexesWas: 3,
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1767198899, i: 3 }),
    signature: {

```

```

mongosh mongodb+srv://<credentials>@cluster0.eay9jl.mongodb.net/
insertedId: ObjectId('6a867cb73c882e93d293a169')
}
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.createIndex({ major: 1 });
major_1
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.getIndexes();
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { major: 1 }, name: 'major_1' }
]
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.createIndex({ major: 1, age: -1 });
major_1_age_-1
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.dropIndex("major_1");
{
  nIndexesWas: 3,
  ok: 1,
  $clusterTime: {
    clusterTime: Timestamp({ t: 1787198809, i: 3 }),
    signature: {
      hash: Binary.createFromBase64('KgMI8JFKdVdg5T5x5zEO1P6RV08=', 0),
      keyId: Long('7616446989839892482')
    }
  },
  operationTime: Timestamp({ t: 1787198809, i: 3 })
}
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.aggregate([
  { $group: { _id: "$major", total: { $sum: 1 } } }
]);
[
  { _id: 'Mathematics', total: 2 },
  { _id: 'Computer Science', total: 2 },
  { _id: 'Physics', total: 1 }
]
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.aggregate([
  { $group: { _id: "$major", avgAge: { $avg: "$age" } } }
]);
[
  { _id: 'Mathematics', avgAge: 22 },
  { _id: 'Physics', avgAge: 23 },
  { _id: 'Computer Science', avgAge: 21 }
]
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.aggregate([
  { $sort: { age: -1 } }
]);
[
  {
    _id: ObjectId('6a867ad63c882e93d293a166'),
    name: 'Charlie',
    age: 23,
  }
]

```

```

mongosh mongodb+srv://<credentials>@cluster0.eay9jl.mongodb.net/
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.aggregate([
  { $sort: { age: -1 } }
]);
[
  {
    _id: ObjectId('6a867ad63c882e93d293a166'),
    name: 'Charlie',
    age: 23,
    major: 'Physics'
  },
  {
    _id: ObjectId('6a867ad63c882e93d293a165'),
    name: 'Bob',
    age: 22,
    major: 'Mathematics'
  },
  {
    _id: ObjectId('6a867ad63c882e93d293a168'),
    name: 'Eve',
    age: 22,
    major: 'Mathematics'
  },
  {
    _id: ObjectId('6a867ad63c882e93d293a167'),
    name: 'David',
    age: 21,
    major: 'Computer Science'
  },
  {
    _id: ObjectId('6a867cb73c882e93d293a169'),
    name: 'Alice',
    age: 21,
    major: 'Computer Science'
  }
]
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.courses.insertMany([
  { course: "DBMS", major: "Computer Science" },
  { course: "Algebra", major: "Mathematics" },
  { course: "Quantum Physics", major: "Physics" }
]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a867eb23c882e93d293a16a'),
    '1': ObjectId('6a867eb23c882e93d293a16b'),
    '2': ObjectId('6a867eb23c882e93d293a16c')
  }
}

```

```

mongosh mongodb+srv://<credentials>@cluster0.eay9yl.mongodb.net/
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.courses.insertMany([
  { course: "DBMS", major: "Computer Science" },
  { course: "Algebra", major: "Mathematics" },
  { course: "Quantum Physics", major: "Physics" }
]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a867eb23c882e93d293a16a'),
    '1': ObjectId('6a867eb23c882e93d293a16b'),
    '2': ObjectId('6a867eb23c882e93d293a16c')
  }
}
Atlas atlas-4upy6x-shard-0 [primary] schoolDB> db.students.aggregate([
  { $lookup: {
    from: "courses",
    localField: "major",
    foreignField: "major",
    as: "enrolledCourses"
  } }
]);
[
  {
    _id: ObjectId('6a867ad63c882e93d293a165'),
    name: 'Bob',
    age: 22,
    major: 'Mathematics',
    enrolledCourses: [
      {
        _id: ObjectId('6a867eb23c882e93d293a16b'),
        course: 'Algebra',
        major: 'Mathematics'
      }
    ]
  },
  {
    _id: ObjectId('6a867ad63c882e93d293a166'),
    name: 'Charlie',
    age: 23,
    major: 'Physics',
    enrolledCourses: [
      {
        _id: ObjectId('6a867eb23c882e93d293a16c'),
        course: 'Quantum Physics',
        major: 'Physics'
      }
    ]
  }
],

```

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major: 'Physics',
enrolledCourses: [
  {
    _id: ObjectId('6a867eb23c882e93d293a16c'),
    course: 'Quantum Physics',
    major: 'Physics'
  }
],
{
  _id: ObjectId('6a867ad63c882e93d293a167'),
  name: 'David',
  age: 21,
  major: 'Computer Science',
  enrolledCourses: [
    {
      _id: ObjectId('6a867eb23c882e93d293a16a'),
      course: 'DBMS',
      major: 'Computer Science'
    }
  ]
},
{
  _id: ObjectId('6a867ad63c882e93d293a168'),
  name: 'Eve',
  age: 22,
  major: 'Mathematics',
  enrolledCourses: [
    {
      _id: ObjectId('6a867eb23c882e93d293a16b'),
      course: 'Algebra',
      major: 'Mathematics'
    }
  ]
},
{
  _id: ObjectId('6a867cb73c882e93d293a169'),
  name: 'Alice',
  age: 21,
  major: 'Computer Science',
  enrolledCourses: [
    {
      _id: ObjectId('6a867eb23c882e93d293a16a'),
      course: 'DBMS',
      major: 'Computer Science'
    }
  ]
}
]

```